

Inclusive Web Platform

Implementation Progress

A practical digital bridge connecting persons with disabilities to real hospitality jobs, training, employers, and support.

Zaidan Mansour | Charbel Rizk

Supervisors: Dr. Jean Yves Youssef | Dr. Roger Obeid | Dr. Rima Kilany

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PROGRESS REVIEW

**Connect people.
Support decisions.
Keep humans in control.**

A bridge to real opportunities - not a replacement for people

Technology organizes access and communication; employers, candidates, families, and experts make the decisions.

WHAT EXISTS

Real people and real hospitality businesses

Persons with disabilities
Families and associations
Employers and training providers
Join inclusion expertise

THE PLATFORM

Makes connection easier and clearer

Shows opportunities
Structures abilities and tasks
Explains compatibility
Supports communication

HUMAN CONTROL

People remain responsible for every outcome

The platform does not invent jobs.
It does not diagnose disability.
It does not automatically hire or reject.
It helps people approach businesses.

Functional and non-functional requirements

The engineering scope includes both user capabilities and measurable quality constraints.

Functional requirements

- + Secure accounts and authorized verification
- + Candidate, company, and employer profiles
- + AI-assisted editable profile workflow
- + Explainable matching and applications
- + Employer, verifier, and admin operations
- + Keyboard and optional voice navigation

Non-functional requirements

- + WCAG 2.1 AA target and responsive design
- + Security, privacy, consent, and role isolation
- + English, French, Arabic, and RTL behavior
- + Transparent and auditable decisions
- + Reliability, automated tests, and monitoring
- + Measured response-time and performance improvement

Implemented feature summary

The current prototype already connects the four user roles through protected and accessible workflows.

AI profile builder

Voice or text -> editable transcript -> consent -> reviewable suggestions

Candidate verification

Email confirmation plus private disability-card review by an authorized verifier

Security controls

JWT session, server-confirmed role, protected routes and APIs, private storage

Matching + applications

Deterministic task score, explanations, applications, and stored outcomes

Employer tools

Company profile, vacancy creation, requirements, important tasks, assistance

Admin + verifier

Users, applications, candidate evidence decisions, and dataset catalogue

Accessible navigation

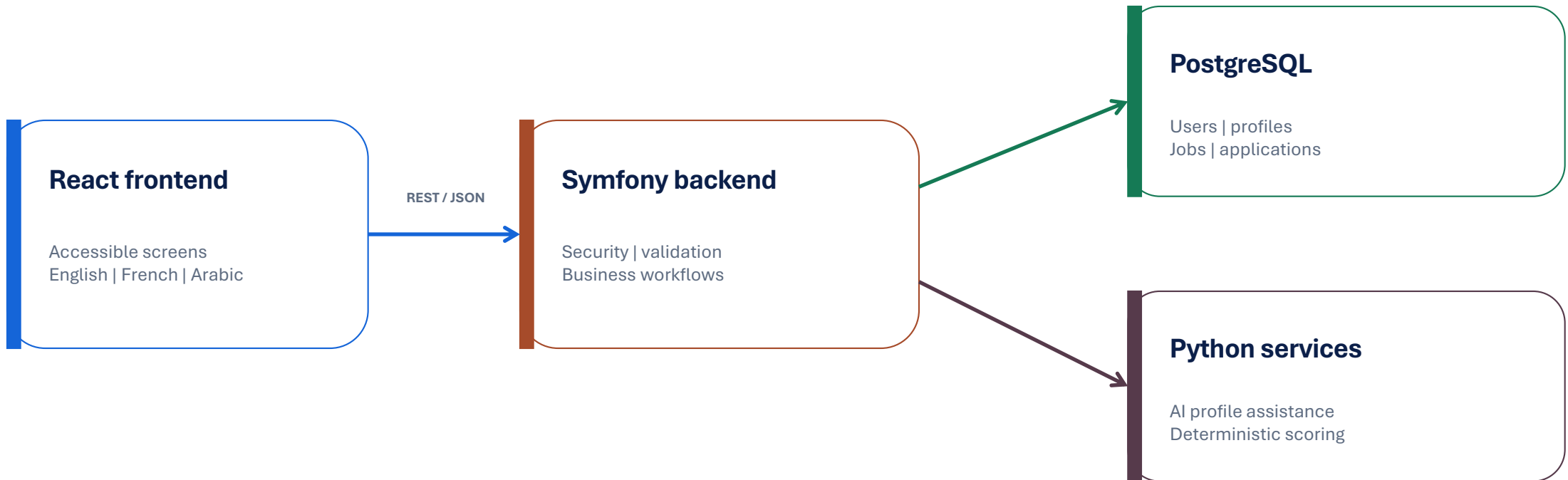
Keyboard, arrow-key support, visible focus, status announcements, optional voice

Multilingual UI

English, French, Arabic, RTL layout, stable codes, and locale tests

Architecture and security boundaries

The browser never grants authority; Symfony validates identity, role, data, and every protected operation.



Trust boundary: Symfony confirms the session and role, protects private evidence, validates input, and controls calls to both Python services.

Why deterministic scoring instead of machine learning?

The current evidence is a controlled expert catalogue, not a labelled history of successful and unsuccessful hires.

CHOSEN: VERSIONED RULES

Task score

$$\text{Sum}(\text{adjusted weight} \times \text{feasibility factor}) / \text{Sum}(\text{adjusted weight}) \times 100$$

Feasibility factor: 1.00 feasible | 0.75 with offered support | 0.00 avoid

Highlighted task: weight x 1.5

Combined result: tasks + practical abilities (25%) + education (10%) when configured

Required education/ability and mandatory unsafe tasks remain explicit eligibility gates

Deterministic rules

- + Uses expert task assessments
- + Every factor is visible
- + Reproducible and auditable
- + Works with the current controlled dataset

Machine learning now

- + Needs representative labelled outcomes
- + May reproduce historical hiring bias
- + Harder to explain individual factors
- + No suitable training evidence currently exists

Why i18next with react-i18next?

The interface changes language at runtime while IDs, roles, permissions, and scoring inputs remain stable.

Custom React context

NOT SELECTED

Few dependencies, but we would rebuild fallback, plural rules, namespaces, formatting, and missing-key checks.

Duplicated pages

NOT SELECTED

Simple for one screen, but triples maintenance, creates translation drift, and complicates routing and testing.

i18next + React

SELECTED

Runtime switching, namespaces, fallback, interpolation, plural rules, React hooks, and gradual route migration.

Implemented: 3 locales x 7 namespaces | Arabic RTL | English fallback | stable codes/IDs | locale parity tests

Why cloud Whisper-1 and OpenAI services?

The current prototype prioritizes multilingual quality and rapid delivery without maintaining local speech-model infrastructure.

Browser speech API

REJECTED

Low backend work, but browser support, language quality, behavior, and availability vary by device.

Self-hosted Whisper

DEFERRED

More infrastructure control and offline potential, but requires CPU/GPU capacity, model operations, updates, and load testing.

OpenAI whisper-1

SELECTED NOW

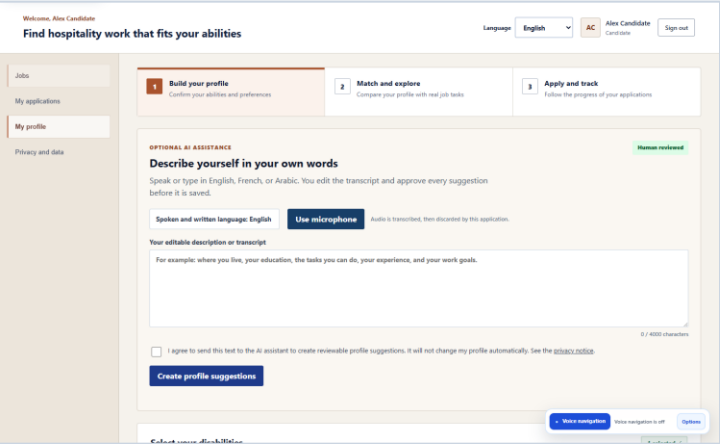
Multilingual transcription, consistent API, no model hosting, and a replaceable service boundary. Trade-offs: network, cost, and latency.

Current controlled flow

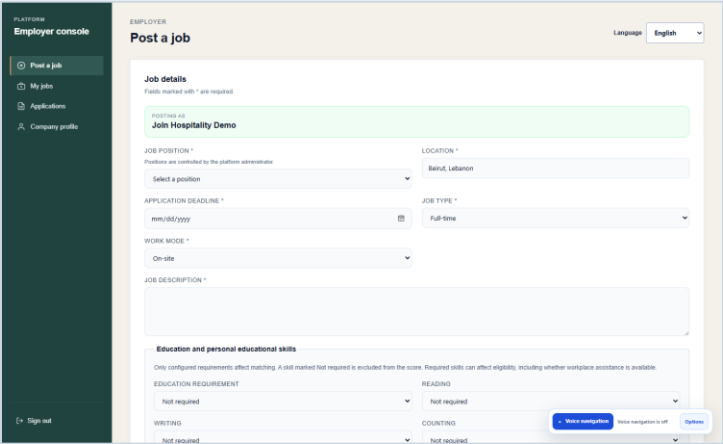
Audio -> whisper-1 transcript -> schema-constrained interpretation -> deterministic registry authorization -> permitted UI action -> cloud TTS or browser fallback

Running workflows and automated verification

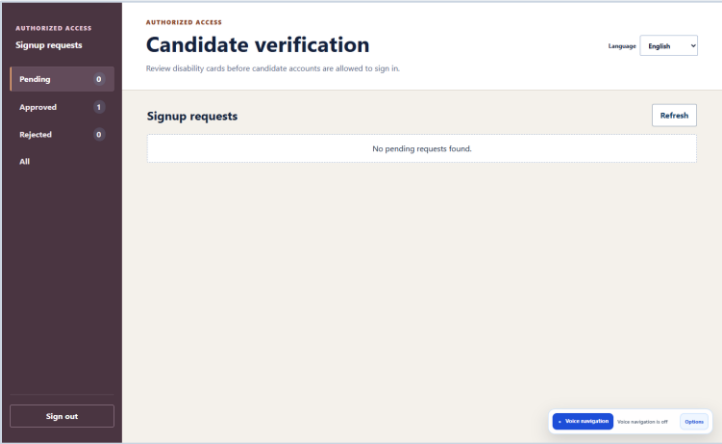
The screens are connected to protected backend services and supported by repeatable tests.



AI profile + consent



Employer requirements



Authorized verification

23 / 23

API integration checks

26 / 26

Browser E2E scenarios

17 / 17

Scoring unit tests

3 x 7

Locales x namespaces

What we are improving now

Current work focuses on stabilization, interface quality, accessibility retesting, catalogue semantics, and richer profiles.

Stability + security

- Resolve workflow and API issues
- Retest wrong-role and anonymous access
- Improve validation, errors, and notifications

Design + accessibility

- Modernize role dashboards
- Improve responsive flow and visual hierarchy
- Retest keyboard, zoom, screen reader, and RTL

Catalogue + profiles

- Validate 7 active -> 15 expected roles
- Preserve Personal Education hierarchy
- Improve candidate and company information

Manual evidence still required: native 200% zoom, documented NVDA/browser pairing, and representative-user validation.

Reduce AI, voice-navigation, and page response time

The objective is measurable improvement in both actual latency and the waiting experienced by the user.

1. Measure first

Timestamp recording, upload, transcription, classification, registry/action, TTS, API, query, and render stages. Report p50 and p95 before/after.

2. Optimize AI path

Trim silence and payloads, bound page context/history, cancel stale requests, keep clients warm, use deterministic shortcuts safely, and keep speech fallback.

3. Optimize platform

Profile database queries, paginate large data, keep route splitting, optimize assets, add indexes, and cache only safe repeated data when measurements justify it.

Performance claim only after repeatable baseline, change, retest, and documented result.

Remaining product work and demonstration

The operational core remains; the next phase broadens access, improves employer support, and prepares validated delivery.

REMAINING PRODUCT WORK

- 1 Candidate chooses Job, Training, or Both
- 2 Visitors browse jobs, training, and companies safely
- 3 Employer records adaptation / assistance for each task
- 4 Upgrade company, candidate, and employer profiles
- 5 Complete and validate the 15-role catalogue
- 6 Finish accessibility evidence, monitoring, capacity tests, and deployment

5-MINUTE DEMO

00:00 Candidate profile + consent
01:00 Match score + explanation
02:00 Employer requirements + assistance
03:00 Admin catalogue
04:00 Language + keyboard + protected route

The platform connects people to real businesses and supports human decisions instead of replacing them.